The Russo-Ukrainian War

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Research Questions

- What are the **temporal and geographic patterns** of Russian missile and UAV strikes?
- Which regions are **most heavily targeted**, and why?
- Can we **predict fatality presence** based on strike characteristics?
- What factors **drive casualty counts** when fatalities occur?
- How can these insights inform **humanitarian response and defense strategy**?

Data & Methodology

Primary Datasets (2022-2026):

1. **ACLED** (Armed Conflict Location & Event Data)
   - 6,119 conflict events
   - 10,000+ fatalities
   - Geolocation data (lat/long)
   - Civilian targeting indicators

1. **Missile/UAV Attack Data** (Kaggle)
   - 85,814 recorded airstrikes
   - Launch and destruction counts by model
   - Temporal records spanning 4 years

Approach:

- Exploratory Data Analysis (EDA)
- Temporal trend analysis
- Geospatial clustering (K-Means, k=5)
- Feature engineering (temporal, geospatial)
- Predictive modeling (Lasso Regression, Random Forest, Logistic Regression)

Key Statistics

Scale of the Conflict

- **6,119** conflict events recorded
- **10,000**+ total fatalities
- **85,814** total recorded airstrikes
- **1.**7 average fatalities per strike
- **1,030** unique locations affected

Weapon Systems

- **Shahed-136/131** accounts for **88%** of all airstrikes
- Iranian-origin UAV dominates attack patterns
- **70%**+ overall destruction rate for launched projectiles
- **50%**+ of events result in zero fatalities

Geographic Concentration

Most Affected Regions:

1. **Donetsk** - 2,460+ strikes (most heavily targeted)
2. **Kharkiv** - 790+ strikes
3. **Zaporizhia** - 760+ strikes

Key Finding: Eastern frontline regions (Donetsk, Luhansk) experience disproportionately high attack concentrations, correlating with greater casualty counts despite not representing the majority of total incidents.

Donetsk and Luhansk alone account for a **significant percentage of total conflict-related fatalities**.

Temporal Trends

Attack intensity remained relatively stable through mid-2024, then escalated sharply with:

- **Peak period:** July 2025
- **Sharp escalation:** Mid-2024 onwards
- **Recent decline:** Early 2026

This temporal pattern suggests:

- Shifts in military strategy
- Seasonal variations
- Response to defensive improvements
- Correlation with broader conflict phases

Lasso Regression for Feature Selection

Task: Identify Key Predictors of Fatality Magnitude

Purpose: Determine which features most strongly influence the number of fatalities when casualties occur.

Key Findings:

- **Geospatial clusters** show strong positive correlation with fatality counts
- **Population density** of target areas associated with increased casualties
- **Temporal features** (day of year, month) indicate seasonal fatality variations
- **Geographic coordinates** (longitude/latitude) are critical predictors

Interpretation: When strikes do cause fatalities, location and temporal context significantly influence the scale of impact. These features were selected as most important for our binary classification model.

Predictive Modeling Results

Random Forest Classifier

Task: Predict presence of fatalities (binary classification)

Performance:

- **Accuracy: 0.90**
- **Precision: 0.89**
- **Recall: 0.90**
- **F1-Score: 0.90**

Top Predictive Features:

1. Day of year
2. Longitude/Latitude of target
3. Population of target city
4. Temporal patterns (month, day of week)

Model Insights

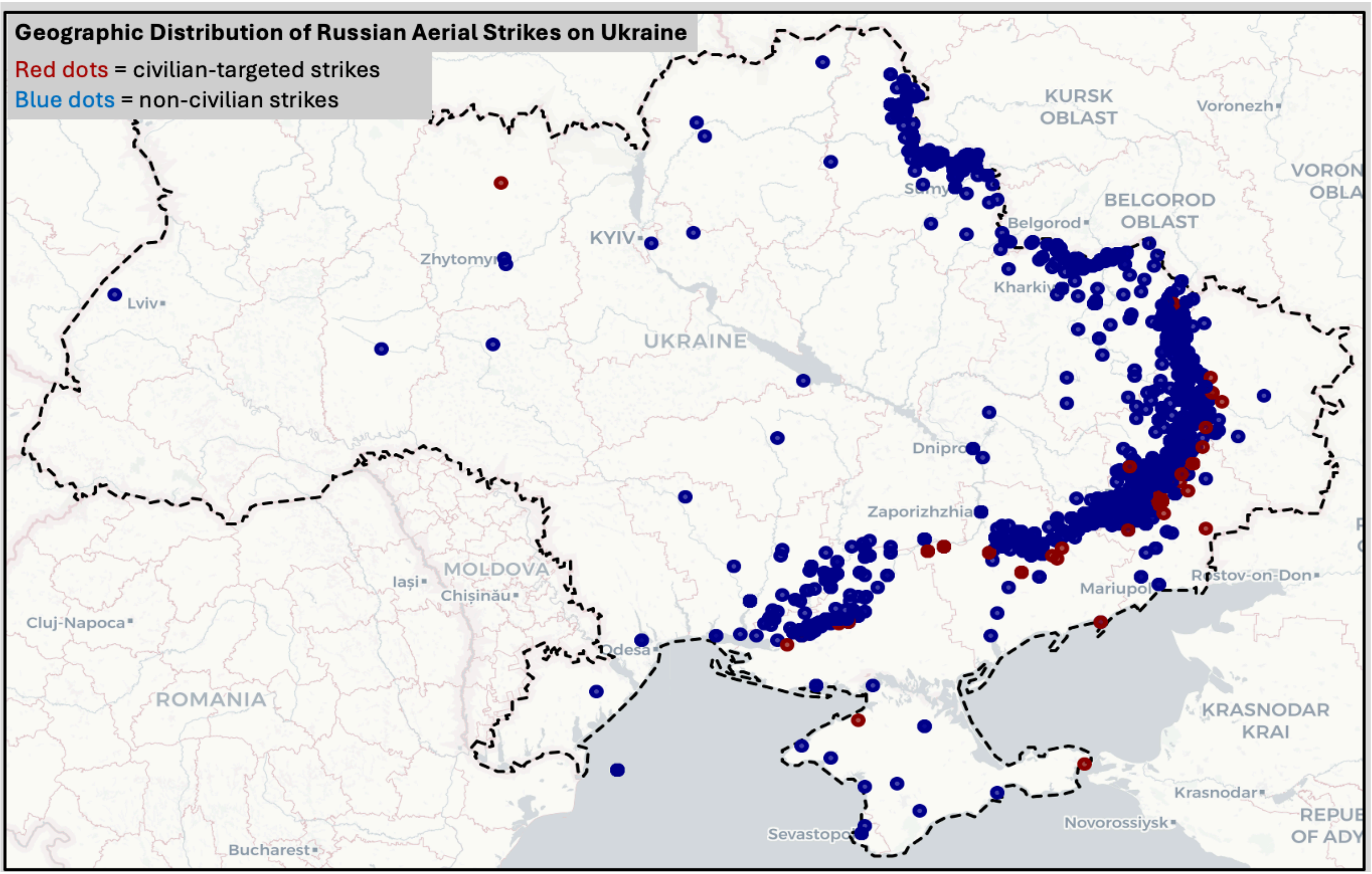
Key Finding: Geographic and temporal factors are **crucial determinants** of strike severity and casualty patterns.

What the Model Tells Us

- **Location matters most** - Specific coordinates predict fatality likelihood with high accuracy
- **Population density is critical** - Strikes on populated areas are more likely to cause casualties
- **Seasonal patterns exist** - Certain times of year show higher fatality correlation
- **Predictability is high** - 90% accuracy suggests systematic, non-random targeting patterns

This predictive capability offers a powerful tool for understanding and potentially mitigating the human cost of these attacks.

Geographic Distribution Map



Interactive 3D Visualization

Explore the conflict in detail:

Access Interactive 3D Map

Features:

- Regional attack intensity (color-coded)
- Individual strike fatalities (3D bars)
- Hover tooltips with detailed statistics
- Rotatable 3D view to explore hotspots

This visualization allows stakeholders to understand both the scale and distribution of attacks across Ukraine.

Implications for Stakeholders

Governments & Military Leadership

- Evaluate effectiveness of current air-defense strategies
- Identify regions where strategic changes could reduce civilian harm
- Allocate resources to high-risk areas

Humanitarian Organizations

- Target aid and protection efforts to most vulnerable regions
- Understand casualty patterns for response planning
- Provide evidence-based support recommendations

Policy Makers & International Community

- Quantitative evidence on conflict scale and impact
- Support decisions on aid allocation and sanctions
- Inform diplomatic and military strategies

Accountability Mechanisms

- Document targeting patterns for future accountability
- Analyze civilian vs. military infrastructure targeting
- Contribute to conflict documentation efforts

Ethical & Societal Implications

This research prioritizes **preservation of human life** as the fundamental objective.

Potential Applications:

- Guide investments in local air-defense technology
- Inform evacuation and protection strategies for at-risk populations
- Contribute to humanitarian resource allocation
- Support evidence-based policy decisions

Our Commitment:

- Transparent methodology and data sources
- Acknowledgment of data limitations
- Respect for victims and affected communities
- Contribution to understanding conflict dynamics for peace-building

Limitations & Caveats

- **ACLED data** relies on media reporting; coverage gaps may exist in isolated areas
- **Missile/UAV data** sourced from Ukrainian media; reflects Ukrainian reporting
- **Geolocation accuracy** varies; some events lack precise coordinates
- **Temporal lag** between incident occurrence and data publication
- **Zero-fatality events** may be underreported in media sources
- **Model predictions** assume historical patterns continue; future patterns may differ

Future Work

Potential Extensions:

- Integrate additional data sources (economic indicators, social media sentiment)
- Develop spatio-temporal deep learning models
- Predict specific types of casualties or infrastructure damage
- Analyze economic impact and reconstruction needs
- Model trajectory of conflict escalation/de-escalation

Next Steps:

- Refine geospatial prediction models
- Develop early warning systems for escalating violence
- Expand analysis to include secondary effects of attacks

Project Resources

Full Analysis & Code:

GitHub Repository

Includes:

- Complete Jupyter notebooks with analysis pipeline
- Data preprocessing and feature engineering code
- Model training and evaluation scripts
- All EDA visualizations
- Interactive map application code
- Poster and presentations

Questions?

Interactive Map: https://ukraine-3d-map-nt7et2c3m46k2boo2vqnfd.streamlit.app

GitHub Repository: https://github.com/lukef533/Russo-Ukrainian-Airstrike-Analysis